

Bridge Course for Engineering Chemistry

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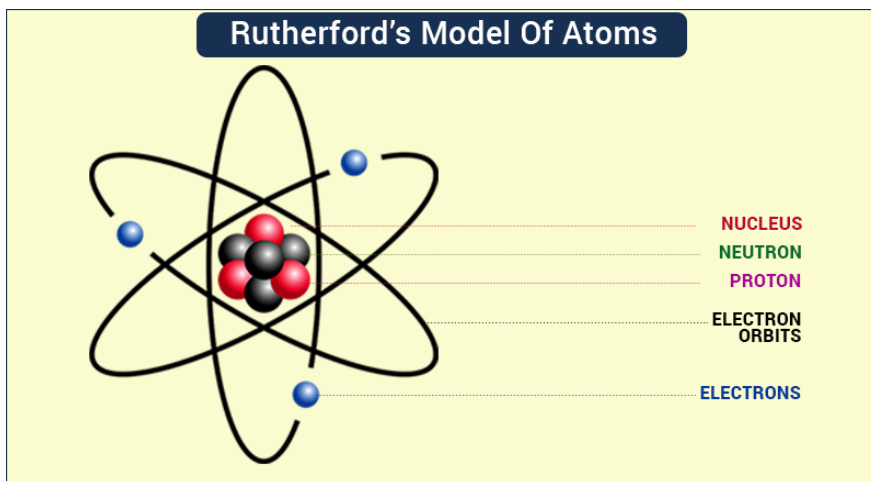
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Module I: Atomic Structure

Sir Ernest Rutherford, the New Zealand-born physicist described **the famous atomic model** or **planetary model of the atom** in 1911. According to the model the atom is described as a tiny, dense, positively charged core called a nucleus, in which nearly all the mass is concentrated, around which the light, negative constituents, called electrons, circulate at some distance, much like planets revolving around the Sun.



Rutherford Atomic Model Experiment

In Rutherford's experiment, he bombarded high energy streams of α -particles on a thin gold foil of 100 nm thickness. The streams of α -particles were directed from a radioactive source. He conducted the experiment to study the deflection produced in the trajectory of α -particles after interaction with the thin sheet of gold. To study the deflection, he placed a screen made up of zinc sulfide around the gold foil.

On the basis of the observations made during the experiment, Rutherford concluded that

- Major space in an atom is empty – A large fraction of α -particles passed through the gold sheet without getting deflected. Therefore, the major part of an atom must be empty.
- The positive charge in an atom is not distributed uniformly and it is concentrated in a very small volume – Few α -particles when bombarded were deflected by the gold sheet. They were deflected minutely and at very small angles. Therefore he made the above conclusion.
- Very few α -particles had deflected at large angles or deflected back. Moreover, very few particles had deflected at 180° . Therefore, he concluded that the positively charged particles covered a small volume of an atom in comparison to the total volume of an atom.

Postulates of Rutherford atomic model based on observations and conclusions

- An atom is composed of positively charged particles. Majority of the mass of an atom was concentrated in a very small region. This region of the atom was called as the **nucleus** of an atom. It was found out later that the very small and dense nucleus of an atom is composed of neutrons and protons.
- Atoms nucleus is surrounded by negatively charged particles called **electrons**. The electrons revolve around the nucleus in a fixed circular path at very high speed. These fixed circular paths were termed as “**orbits**.”
- An atom has no net charge or they are **electrically neutral** because electrons are negatively charged and the densely concentrated nucleus is positively charged. A strong electrostatic force of attractions holds together the nucleus and electrons.
- The size of the nucleus of an atom is very small in comparison to the total size of an atom.

Limitations of Rutherford Atomic Model

Rutherford’s experiment was unable to explain certain things. They are:

- Rutherford’s model was unable to explain the stability of an atom. According to Rutherford’s postulate, electrons revolve at a very high speed around a nucleus of an atom in a fixed orbit. However, Maxwell explained accelerated charged particles release electromagnetic radiations. Therefore, electrons revolving around the nucleus will release electromagnetic radiation.
- The electromagnetic radiation will have energy from the electronic motion as a result of which the orbits will gradually shrink. Finally, the orbits will shrink and collapse in the nucleus of an atom. According to the calculations, if Maxwell’s explanation is followed Rutherford’s model will collapse with 10^{-8} seconds. Therefore, Rutherford atomic model was not following Maxwell’s theory and it was unable to explain an atom’s stability.
- Rutherford’s theory was incomplete because it did not mention anything about the arrangement of electrons in the orbit. This was one of the major drawbacks of Rutherford atomic model.

Bohr’s theory to hydrogen-like atoms and ions

The objections of Rutherford atomic model was contested by Niels Bohr atomic model in 1913. He proposed the quantum theory of an atom. The theory was based on the quantum theory of radiation. Bohr retained the main postulates of Rutherford planetary model and did some modification on the basis of quantum physics. Hence, Bohr atomic model is also known as Rutherford-Bohr atomic model. He depicts the atom as a tiny, spherical body which consists nucleus at center and negatively charged particles (electrons) revolving around nucleus in a certain path known as orbit. He proposed some new postulate with same basis concepts of Rutherford theory.

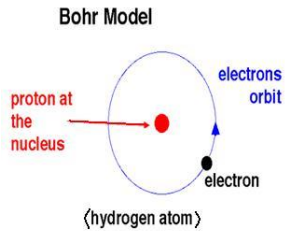
Bohr's Atomic Model

In order to explain the stability of an atom, Neils Bohr gave a new arrangement of electrons in the atom in 1913. According to Neils Bohr, the electrons could revolve around the nucleus in only '**certain orbits**' (energy levels), each orbit having a different radius. When an electron is revolving in a particular orbit or particular energy level around the nucleus, the electron does not radiate energy (lose energy) even though it has accelerated motion around the nucleus.

Postulates of Niels Bohr Atomic Theory

The important postulates in his theory are,

1. Electrons revolve round the nucleus with definite velocities in concentric circular orbits situated at definite distances from the nucleus. The energy of an electron in a certain orbit remains constant. As long as it remains in that orbit, it neither emits nor absorbs energy. These are termed stationary states or main energy states.
2. Bohr proposed that the angular momentum of an electron is quantized. Thus, the motion of an electron is restricted to those orbits where its angular momentum is an integral multiple of $h/2\pi$, where h is Planck's constant.
3. Thus we have the relationship $mvr = nh/2\pi$, where m is mass of electron, v is velocity of electron of said orbit, r is radius of that orbit, n is a simple integer.
4. The stationary states or allowed energy levels are only those where $n = 1, 2, 3, \dots$. This is called Bohr quantum condition.
5. The energy of an electron changes only when it moves from one orbit to another. An electronic transition from an inner orbit to outer orbit involves absorption of energy. Similarly, when an electron jumps from an outer orbit to inner orbit it releases energy, which is equal to the difference between the two energy levels.
6. The energy thus released in the form of a radiation of a certain frequency appears in the form a line in the atomic spectrum. If the energy of an electron in the outer orbit (n_2) is E_2 and energy of electron in the inner orbit (n_1) is E_1 then $E_2 - E_1 = \Delta E = h\nu$.
7. The value of n could be small integers 1, 2, 3 and these correspond to the first, second, third, and so on. Quantum states are shells for the electron; n is termed as principal quantum number.
8. Based on the Bohr theory, he calculated the radii of the various orbits and the energies associated with the electrons present in those shells.



Bohr Model of Hydrogen

Bohr used the concept of quantization of energy and laws of classical physics. He gave many mathematical expressions like radius of orbits, velocity and energy of electrons. These expressions satisfactorily explain the spectrum of hydrogen. Let's calculate all these expressions to explain the spectrum of hydrogen (${}^1\text{H}^1$).

1. Radius of nth orbit

According to Bohr model, the attraction force between electron and nucleus is balanced by centrifugal force of electron which is due to motion of electron and tend to take electron away from nucleus.

Let's take,

Atomic number of atom = Z

Charge on electron = e

Charge on nucleus = Ze

Radius of nth orbit = r_n

The electrostatic force of attraction = Ze^2/r_n^2 -----(1)

The centripetal force = mV^2/r_n -----(2)

Since both forces balanced each other. Hence from equation 1 and 2

$$Ze^2/r_n^2 = mV^2/r_n$$

or,

$$V^2 = Ze^2/r_nm$$
 -----(3)

Form Bohr postulate $mVr_n = nh/2\pi$

$$\text{or } m^2V^2r_n^2 = n^2h^2/4\pi^2$$

Putting the value of V^2 from eqn----(3)

$$m^2 \times Ze^2/r_nm \times r_n^2 = n^2h^2/4\pi^2$$

or,

$$m \times Ze^2 \times r_n = n^2h^2/4\pi^2$$

or,

$$r_n = n^2 h^2 / 4\pi^2 m Z e^2 \text{ -----(4)}$$

Since

$$h = 6.62 \times 10^{-27} \text{ erg.sec (Plank's Constant)}$$

$$\pi = 3.142$$

$$m = 9.109 \times 10^{-28} \text{ gm}$$

$$e = 4.803 \times 10^{-10} \text{ esu}$$

so

$$h^2 / 4\pi^2 m e^2 = 0.529 \text{ \AA}$$

Putting the value of constants in (4)

$$r_n = n^2 / Z \times 0.529 \text{ \AA}$$

- Since the atomic number for hydrogen is one, so, the radius of nth orbit of hydrogen will be $r_1 = n^2 \times 0.529 \text{ \AA}$.
- Since the value of Z is constant for an atom, $r_n \propto n^2$, so radius increases with increasing the value of n.
- If the value of n is constant, $r_n \propto 1/Z$
- Hence, radius of a particular orbit decreases with increasing the atomic number.

2. Calculation of the velocity of electron in Bohr orbit

Form Bohr postulate

$$Z e^2 / r_n^2 = m V^2 / r_n$$

Or,

$$V^2 = Z e^2 / r_n m$$

$$\text{Since } r_n = n^2 h^2 / m Z e^2 4\pi^2$$

$$\text{Hence } V^2 = Z^2 e^4 4\pi^2 / n^2 h^2$$

Or,

$$V = Z e^2 2\pi / n h \text{ -----(5)}$$

Putting all constants (e, r, h, π) values in equation (5)

$$V = Z / n \times 2.188 \times 10^8 \text{ cm/sec}$$

For hydrogen atom, $Z = 1$

$$\text{Hence } V = 2.188 \times 10^8 / n \text{ cm.sec}^{-1}$$

3. Energy of electron in n^{th} orbit

According to Bohr atomic model, the maximum energy value of electron at infinite is zero because of negligible attraction force between electron and nucleus at infinite distance. Hence, as electron comes closer to nucleus, the energy becomes negative.

Energy of electron is the sum of its potential energy because the electron lies in the field of the positive nucleus and kinetic energy which is due to motion of electron. The potential energy of electron is negative and equals to $-Ze^2/r$, while the kinetic energy is positive and equals to $1/2mv^2$.

Hence the total energy of electron

$$E_n = -Ze^2/r + 1/2mv^2$$

$$\text{Since } mv^2 = Ze^2/r_n$$

$$\text{Hence } E_n = -(1/2Ze^2 \times mZe^2 4\pi^2)/n^2 h^2$$

$$E_n = -12Ze^2/r_n$$

We know that

$$r_n = n^2 h^2 / mZe^2 4\pi^2$$

$$\begin{aligned} \text{So } E_n &= (1/2Ze^2 \times mZe^2 4\pi^2)/n^2 h^2 \\ &= -(Z^2 \times 2\pi^2 me^2)/n^2 h^2 \end{aligned}$$

Since $2\pi^2 me^2/h^2$ is a constant value, which is equals to 13.60 eV/atom.

$$\text{So } E_n = -Z^2/n^2 \times 13.60 \text{ eV/atom}$$

$$E_n = -Z^2/n^2 \times 2.179 \text{ erg/atom}$$

$$E_n = -Z^2/n^2 \times 313.6 \text{ KCal/mol}$$

$$E_n = -Z^2/n^2 \times 21.79 \times 10^{-19} \text{ J/atom}$$

For hydrogen $Z = 1$ so for

$$\text{So } E_n = -1/n^2 \times 13.60 \text{ eV/atom}$$

Spectrum of Hydrogen Atom

Bohr atomic model states that the spectrum series arises when an electron jumps from an initial stationary orbit (with principle quantum number n_i) to the final orbit (with principle quantum

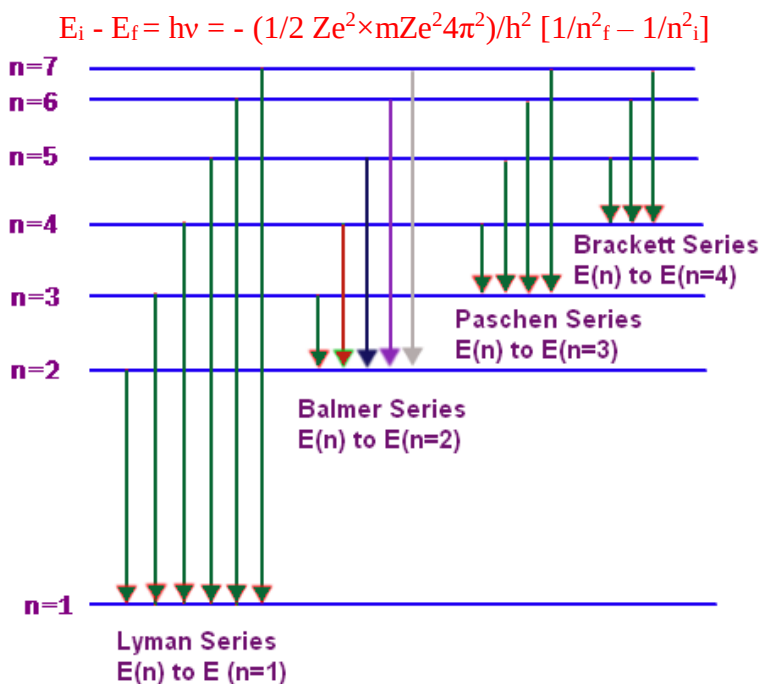
number n_f). Hence, the energy changes associated with these orbits are emitted as photon of frequency μ .

$$\Delta E = h\nu = E_i - E_f$$

Since $E = -1/2 Ze^2/r_n$

and

$$r_n = n^2 h^2 / mZe^2 4\pi^2$$



The Four Electronic Quantum Numbers

Quantum numbers designate specific shells, subshells, orbitals, and spins of electrons. This means that they describe completely the characteristics of an electron in an atom, i.e., they describe each unique solution to the Schrödinger equation, or the wave function, of electrons in an atom. There are a total of four quantum numbers: the principal quantum number (n), the orbital angular momentum quantum number (l), the magnetic quantum number (m_l), and the electron spin quantum number (m_s). The principal quantum number, n , describes the energy of an electron and the most probable distance of the electron from the nucleus. In other words, it refers to the size of the orbital and the energy level an electron is placed in. The number of subshells, or l , describes the shape of the orbital. It can also be used to determine the number of angular nodes. The magnetic quantum number, m_l , describes the energy levels in a subshell, and m_s refers to the spin on the electron, which can either be up or down.

The Principal Quantum Number (n)

The principal quantum number, n , designates the principal electron shell. Because n describes the most probable distance of the electrons from the nucleus, the larger the number n is, the farther the electron is from the nucleus, the larger the size of the orbital, and the larger the atom is. n can be any positive integer starting at 1, as $n=1$ designates the first principal shell (the innermost shell). The first principal shell is also called the ground state, or lowest energy state.

The Orbital Angular Momentum Quantum Number (l)

The orbital angular momentum quantum number l determines the shape of an orbital, and therefore the angular distribution. The number of angular nodes is equal to the value of the angular momentum quantum number l . Each value of l indicates a specific s, p, d, f subshell (each unique in shape). The value of l is dependent on the principal quantum number n . Unlike n , the value of l can be zero. It can also be a positive integer, but it cannot be larger than one less than the principal quantum number ($n-1$):

$$l=0,1,2,3,4,\dots,(n-1)$$

The Magnetic Quantum Number (m_l)

The magnetic quantum number m_l determines the number of orbitals and their orientation within a subshell. Consequently, its value depends on the orbital angular momentum quantum number l . Given a certain l , m_l is an interval ranging from $-l$ to $+l$, so it can be zero, a negative integer, or a positive integer.

$$m_l=-l,(-l+1),(-l+2),\dots,-2,-1,0,1,2,\dots,(l-1),(l-2),+l$$

The Electron Spin Quantum Number (m_s)

Unlike n , l , and m_l , the electron spin quantum number m_s does not depend on another quantum number. It designates the direction of the electron spin and may have a spin of $+1/2$, represented by $\uparrow$, or $-1/2$, represented by $\downarrow$. This means that when m_s is positive the electron has an upward spin, which can be referred to as "spin up." When it is negative, the electron has a downward spin, so it is "spin down." The significance of the electron spin quantum number is its determination of an atom's ability to generate a magnetic field or not.

$$m_s=\pm 1/2$$

Example: List the possible combinations of all four quantum numbers when $n=2$, $l=1$, and $m_l=0$.

Solution:

The fourth quantum number is independent of the first three, allowing the first three quantum numbers of two electrons to be the same. Since the spin can be $+1/2$ or $-1/2$, there are two combinations: $n=2$, $l=1$, $m_l=0$, $m_s=+1/2$ and $n=2$, $l=1$, $m_l=0$, $m_s=-1/2$

Subshells

The number of values of the orbital angular number l can also be used to identify the number of subshells in a principal electron shell:

- When $n = 1$, $l = 0$ (l takes on one value and thus there can only be one subshell)
- When $n = 2$, $l = 0, 1$ (l takes on two values and thus there are two possible subshells)
- When $n = 3$, $l = 0, 1, 2$ (l takes on three values and thus there are three possible subshells)

After looking at the examples above, we see that the value of n is equal to the number of subshells in a principal electronic shell:

- Principal shell with $n = 1$ has one subshell

- Principal shell with $n = 2$ has two subshells
- Principal shell with $n = 3$ has three subshells

Therefore:

- Principal shell with $n = 1$ has one s subshell ($l = 0$)
- Principal shell with $n = 2$ has one s subshell and one p subshell ($l = 0, 1$)
- Principal shell with $n = 3$ has one s subshell, one p subshell, and one d subshell ($l = 0, 1, 2$)

We can designate a principal quantum number, n , and a certain subshell by combining the value of n and the name of the subshell (which can be found using l). For example, 3p refers to the third principal quantum number ($n=3$) and the p subshell ($l=1$).

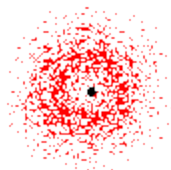
Atomic Orbitals

The atomic orbital is defined as the three dimensional space in the atom where probability of finding electron is maximum. The orbitals are of four types, s, p, d, f. The number of orbitals in a subshell is equivalent to the number of values the magnetic quantum number m_l takes on. A helpful equation to determine the number of orbitals in a subshell is $2l + 1$. This equation will not give you the value of m_l , but the number of possible values that m_l can take on in a particular orbital. For example, if $l=1$ and m_l can have values -1, 0, or +1, the value of $2l+1$ will be three and there will be three different orbitals. The names of the orbitals are named after the subshells they are found in:

	s orbitals	p orbitals	d orbitals
l	0	1	2
m_l	0	-1, 0, +1	-2, -1, 0, +1, +2
Number of orbitals in designated subshell	1	3	5

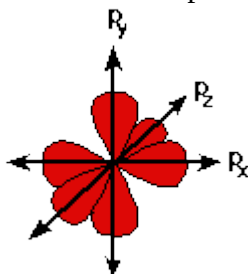
s orbital

Consider a single hydrogen atom: at a particular instant, the position of the electron is plotted. The position is plotted again soon afterward, and it is in a different position. There is no way to tell how it moved from the first place to the second. In the hydrogen case, the electron can be found anywhere within a spherical space surrounding the nucleus.



p orbitals

Not all electrons inhabit s orbitals (in fact, very few electrons live occupy s orbitals). At the first energy level, the only orbital available to electrons is the 1s orbital, but at the second level, as well as a 2s orbital, there are 2p orbitals. A p orbital is shaped like 2 identical balloons tied together at the nucleus. At any one energy level it is possible to have three absolutely equivalent p orbitals pointing mutually at right angles to each other. These are arbitrarily given the symbols p_x , p_y and p_z . This is simply for convenience; the x, y, and z directions change constantly as the atom tumbles in space.



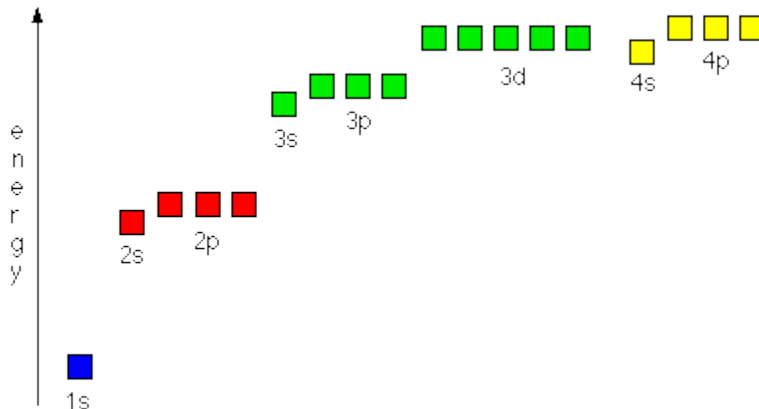
d and f orbitals

In addition to s and p orbitals, there are two other sets of orbitals that become available for electrons to inhabit at higher energy levels. At the third level, there is a set of five d orbitals (with complicated shapes and names) as well as the 3s and 3p orbitals ($3p_x$, $3p_y$, $3p_z$). At the third level there are nine total orbitals. At the fourth level, as well the 4s and 4p and 4d orbitals there are an additional seven f orbitals, adding up to 16 orbitals in all. s, p, d and f orbitals are then available at all higher energy levels as well.

The Aufbau Principle: the order of filling orbitals

Aufbau is a German word meaning building up or construction. Moving from one atom to the next in the periodic table, the electronic structure of the next atom can be determined by fitting an extra electron into the next available orbital. Electrons fill low energy orbitals (closer to the nucleus) before they fill higher energy ones. If there is a choice between orbitals of equal energy, they fill the orbitals singly as far as possible before pairing up.

This filling of orbitals singly where possible is known as Hund's rule. It only applies to orbitals with exactly the same energies (as with p orbitals, for example), and helps to minimize the repulsions between electrons and so makes the atom more stable. The diagram (not to scale) summarizes the energies of the orbitals up to the 4p level.



Hund's Rule

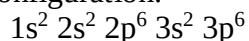
Hund's rules refers to a set of rules that German physicist Friedrich Hund formulated around 1927, which are used to determine the term symbol that corresponds to the ground state of a multi-electron atom.

1. For a given electron configuration, the term with maximum multiplicity has the lowest energy. The multiplicity is equal to $2S+1$, where, S is the total spin angular momentum for all electrons. Therefore, the term with lowest energy is also the term with maximum S .
2. For a given multiplicity, the term with the largest value of the total orbital angular momentum quantum number L has the lowest energy.
3. For a given term, in an atom with outermost subshell half-filled or less, the level with the lowest value of the total angular momentum quantum number J , (for the operator $J=L+S$) lies lowest in energy. If the outermost shell is more than half-filled, the level with the highest value of J is lowest in energy.

Pauli's exclusion principle

It states that in an atom no two electrons can have identical four sets of quantum number, i.e. n , l , m_l may be same but s must be different.

Example: Consider argon's electron configuration:



The exclusion principle asserts that every electron in an argon atom is in a unique state. The $1s$ level can accommodate two electrons with identical n , l , and m_l quantum numbers. Argon's pair of electrons in the $1s$ orbital satisfy the exclusion principle because they have opposite spins, meaning they have different spin quantum numbers, s . One spin is $+\frac{1}{2}$, the other is $-\frac{1}{2}$. (Instead of saying $+\frac{1}{2}$ or $-\frac{1}{2}$ often the electrons are said to be spin-up $\uparrow$ or spin-down $\downarrow$.)

The Heisenberg Uncertainty Principle Schrödinger Wave Equation

Heisenberg stated that the more precisely we can define the position of an electron, the less certainly we are able to define its velocity, and is vice versa. If Δx is the uncertainty in defining the position and Δv the uncertainty in the velocity, the uncertainty principle may be expressed mathematically as:

$$\Delta x \cdot \Delta v \geq \frac{h}{4\pi}$$

Where, h = Plank's constant = 6.626×10^{-34} J s. This implies that it is impossible to know both the position and the velocity exactly.

The concept of an electron following a definite orbit, where its position and velocity are known exactly, must therefore be replaced by the probability of finding an electron in a particular position, or in a particular volume of space. The Schrödinger Wave Equation provides a satisfactory description of an atom in these terms. Solutions to the wave equation are called wave functions and given the symbol ψ . The probability of finding an electron at a point in space whose coordinates are x , y and z is $\psi^2(x, y, z)$.

Concept of Periodic Table

Historical background of the development of Periodic classification

From the very old days, chemists had attempted to classify the elements into some groups based on their chemical and physical properties to systematize the study. At the early days, the chemists had always attempted to find such relationship with the atomic weights of the elements. But as the correct atomic weights, of all the elements known at that time, were not known before 1860, many early attempts failed. Then within a few years (1868-70), the periodic classification taking the atomic weights as the basis came out from the German chemist, Lothar Meyer and the Russian chemist, Mendeleev. Such empirical classifications have now been supported theoretically with the advent of electronic structure of the elements. However, before the attainment of Mendeleev's periodic law, some early attempts which provided the key routes to the thinking of Mendeleev and Lothar Meyer are worth mentioning. Some such historical developments are mentioned below.

Law of Triads

Johann Wolfgang Dobereiner was a chemist in the early 1800s, when the periodic table wasn't in existence, and some chemists were trying to find a type of organizational system for the known elements. Dobereiner noticed a pattern with certain elements that had similar chemical and physical properties. He called these elements triads and, thus, we have **Dobereiner's Law of Triads**. Dobereiner noticed the atomic weight of strontium fell midway between the weights of calcium and barium:

Ca Sr Ba $(40 + 137) \div 2 = 88$
40 88 137

Was this merely a coincidence or did some pattern to the arrangement of the elements exist? Dobereiner noticed the same pattern for the alkali metal triad (Li/Na/K) and the halogen triad (Cl/Br/I).

Li Na K **Cl Br I**
7 23 39 35 80 127

In 1829 Dobereiner proposed the Law of Triads: Middle element in the triad had atomic weight that was the average of the other two members. Soon other scientists found chemical relationships extended beyond triads. Fluorine was added to Cl/Br/I group; sulfur, oxygen, selenium and tellurium were grouped into a family; nitrogen, phosphorus, arsenic, antimony, and bismuth were classified as another group.

Newlands' law of Octave

English chemist John Newlands (1837-1898), having arranged the 62 known elements in order of increasing atomic weights, noted that after interval of eight elements similar physical/chemical properties reappeared. Newlands was the first to formulate the concept of periodicity in the properties of the chemical elements. In 1863 he wrote a paper proposing the Law of Octaves: Elements exhibit similar behavior to the eighth element following it in the table.

Newlands' Arranged Elements in Octaves:

H	F	Cl	Co/Ni	Br	Pd	I	Pt/Ir
Li	Na	K	Cu	Rb	Ag	Cs	Tl
G	Mg	Ca	Zn	Sr	Cd	Ba/V	Pb
Bo	Al	Cr	Y	Ce/La	U	Ta	Th
C	Si	Ti	In	Zn	Sn	W	Hg
N	P	Mn	As	Di/Mo	Sb	Nb	Bi
O	S	Fe	Se	Ro/Ru	Te	Au	Os

Mendeleev's and Lothar Meyer's Periodic Laws

Then in 1869, Russian chemist Dimitri Mendeleev (1834-1907) proposed arranging elements by atomic weights and properties (Lothar Meyer independently reached similar conclusion but published results after Mendeleev). Mendeleev's periodic table of 1869 contained 17 columns with two partial periods of seven elements each (Li-F & Na-Cl) followed by two nearly complete periods (K-Br & Rb-I). In 1871 Mendeleev revised the 17-group table with eight columns (the eighth group consisted of transition elements). This table exhibited similarities not only in small units such as the triads, but showed similarities in an entire network of vertical, horizontal, and diagonal relationships. The table contained gaps but Mendeleev predicted the discovery of new elements. In 1906, Mendeleev came within one vote of receiving the Nobel Prize in chemistry.

This law states that the physical and chemical properties of the elements are the periodic function of their atomic masses. This means that when the elements are arranged in the order of their increasing atomic masses, the elements with similar properties recur at regular intervals. Such orderly recurring properties in a cyclic fashion are said to be occurring periodically.

Merits of Mendeleev's periodic table: Mendeleev's periodic table was one of the greatest achievements in the development of chemistry. Some of the important contributions of his periodic table are given below;

(A) Systematic study of elements: He arranged known elements in order of their increasing atomic masses considering the fact that elements with similar properties should fall in the same vertical column.

(B) Correction of atomic masses: The mendeleev's periodic table could predict errors in the atomic masses of certain elements were corrected. For example, atomic mass of beryllium was corrected from 13.5 to 9. Similarly, with the help of this table, atomic masses of indium, gold, platinum etc. were corrected.

(C) Mendeleev predicated the properties of those missing elements from the known properties of the other elements in the same group. Eka-boron, eka – aluminum and eka – silicon names were given for scandium, gallium and germanium (not discovered at the time of Mandeleev).

(D) Position of noble gases: Noble gases like helium (He), neon (Ne) and argon (Ar) were mentioned in many studies. However, these gases were discovered very late because they are very inert and are present in extremely low concentrations. One of the achievements of

Mendeleev's periodic table was that when these gases were discovered, they could be placed in a new group without disturbing the existing order.

Limitations of Mendeleev's periodic table: In spite of many advantages, the Mendeleev's periodic table has certain defects also. Some of these are given below;

(A) Position of hydrogen: Position of hydrogen in the periodic table is **uncertain**. It has been placed in 1A group with alkali metals, but certain properties of hydrogen resemble those of halogens. So, it may be placed in the group for halogens as well.

(B) Position of isotopes: Isotopes are the atoms of the same element having different atomic masses. Therefore, according to Mendeleev's classification these should be placed at different places depending upon their atomic masses. For example, hydrogen isotopes with atomic masses 1, 2 and 3 should be placed at three places. However, isotopes have not been given separate places in the periodic table because of their similar properties.

(C) Anomalous pairs of elements: In certain pair of elements, the increasing order of atomic masses was not obeyed. In these, Mendeleev placed elements according to similarities in their properties and not in increasing order of their atomic masses.

Modern Periodic Table

Mendeleev's table needed one important modification before it became the modern periodic table – the use of **atomic number** to order the elements. Mendeleev put the elements in order of their relative atomic mass, and this gave him some problems. For example, iodine has a lower relative atomic mass than tellurium, so it should come before tellurium in Mendeleev's table. In order to get iodine in the same group as other elements with similar properties such as fluorine, chlorine and bromine, he had to put it after tellurium, so breaking his own rules. Using **atomic number** instead of **atomic mass** as the organizing principle was first proposed by the British chemist Henry Moseley in 1913, and it solved anomalies like this one. Iodine has a higher atomic number than tellurium.

The periodic table has undergone extensive changes in the time since it was originally developed by Mendeleev and Moseley. Many new elements have been discovered, while others have been artificially synthesized. Each fits properly into a **group** of elements with similar properties. The periodic table is an arrangement of the elements in order of their atomic numbers so that elements with similar properties appear in the same vertical column or group. The figure below shows the most commonly used form of the periodic table. Each square shows the chemical symbol of the element along with its name. Notice that several of the symbols seem to be unrelated to the name of the element: Fe for iron, Pb for lead, etc. Most of these are the elements that have been known since ancient times and have symbols based on their Latin names. The atomic number of each element is written above the symbol.

Group→	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
↓Period																		
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba	* 71 Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra	* 103 Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Uut	114 Fl	115 Uup	116 Lv	117 Uus	118 Uuo
			* 57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb		
			* 89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No		

A period is a horizontal row of the periodic table. There are seven periods in the periodic table, with each one beginning at the far left. A new period begins when a new principal energy level begins filling with electrons. Period 1 has only two elements (hydrogen and helium), while periods 2 and 3 have 8 elements. Periods 4 and 5 have 18 elements. Periods 6 and 7 have 32 elements because the two bottom rows that are separated from the rest of the table belong to those periods. They are pulled out in order to make the table itself fit more easily onto a single page.

A group is a vertical column of the periodic table, based on the organization of the outer shell electrons. There are a total of 18 groups. There are two different numbering systems that are commonly used to designate groups and you should be familiar with both. The traditional system used in the United States involves the use of the letters A and B. The first two groups are 1A and 2A, while the last six groups are 3A through 8A. The middle groups use B in their titles. Unfortunately, there was a slightly different system in place in Europe. To eliminate confusion the International Union of Pure and Applied Chemistry (IUPAC) decided that the official system for numbering groups would be a simple 1 through 18 from left to right. Many periodic tables show both systems simultaneously.

Module II: Concept of Thermodynamics

Thermodynamics is defined as the branch of science that deals with the relationship between heat and other forms of energy, such as work. It is frequently summarized as the three laws of Thermodynamics that describe how different forms of energy can be interconverted.

One of the basic assumptions of thermodynamics is the idea that we can arbitrarily divide the universe into a **system** and its **surroundings**.

A **thermodynamic system** is defined as the quantity of matter or a region in space over which the study of thermodynamics is done. Here quantity of matter may be gas, solid or liquid.

Everything external to the system is called the surroundings or environment. The system is separated from the surroundings by the system **boundary/wall/partition**.

System and surroundings together constitute the universe. Size of the universe depends on the size of the system and surroundings.

The **boundary** between the system and its surroundings can be as real (e.g. the walls of a beaker) or imaginary. A wall may be fixed (e.g. a constant volume reactor), moveable (e.g. a piston), impermeable, permeable, or semi-permeable.

Thermodynamic properties can be divided into 2 (two) categories

1. **Intensive properties:** An intensive property, is a physical property of a system that does not depend on the system size or the amount of material in the system. e.g. density, pressure, temperature etc.
2. **Extensive properties:** An extensive property of a system does depend on the system size or the amount of material in the system. E.g. volume, internal energy are extensive properties.

Thermodynamic Process: A thermodynamic process occurs when the system goes from some initial equilibrium state S1 to a new equilibrium state S2. The system coordinates, eg. T, P etc. will change as a result of the process. The 'path' for the process is defined by a specification of the equation of state.

Reversible and Irreversible process

1. A **reversible process** is a process whose direction can be "reversed" by inducing infinitesimal changes to some property of the system via its surroundings, while not increasing entropy.
2. Throughout the entire reversible process, the system is in thermodynamic equilibrium with its surroundings.

3. Since it would take an infinite amount of time for the reversible process to finish, perfectly reversible processes are impossible. In a *reversible cycle*, the system and its surroundings will be returned to their original states if the forward cycle is followed by the reverse cycle.
4. Irreversible processes, on the other hand can be described as the process where finite changes are made; therefore the system is not at equilibrium throughout the process. At the same point in an irreversible cycle, the system will be in the same state, but the surroundings are permanently changed after each cycle.

Work done in an Isothermal reversible process for expansion of an ideal gas

$$W = nRT \ln(V_2/V_1)$$

Work done in an Adiabatic process : $W_{\text{adia}} = (P_1V_1 - P_2V_2)/(\gamma - 1)$ 2nd law of thermodynamics

Although all natural processes must take place in accordance with the First Law, still it is inadequate for explaining certain behavior of a system, i.e.

- i. Burning of coal
- ii. Flow of water down a hill

Coal burns with O_2 to produce CO_2 and ashes but coal and O_2 cannot be produced by mixing ash and CO_2

Flow of water down a hill is spontaneous but water cannot flow up hill.

Heat Capacity

The amount of heat required to raise the temperature of 1gm of a substance by $1^\circ K$ is called its **specific heat**.

The heat input required to raise the temperature of 1 mole of the substance by $1^\circ K$ is called its **molar heat capacity** or simply **Heat Capacity**. It is denoted by C .

$$C = dq/dT$$

But dq is a path function and its value depends on the actual process followed. So, “at constant pressure” and “at constant volume”-

$$C_V = (dq)_V/dT$$

And

$$C_P = (dq)_P/dT$$

$$\text{But, } dU = (dq)_V$$

And

$$dH = (dq)_P$$

Hence, at constant volume $\Delta U = C_V \Delta T$

and at constant pressure $\Delta H = C_P \Delta T$

C_p is higher than C_v

The heat capacity at constant pressure C_P is greater than the heat capacity at constant volume C_V , because when heat is added at constant pressure, the substance expands and works. When heat is added to a gas at constant volume, we have

$$Q_V = C_V \Delta T = \Delta U + W = \Delta U$$

When heat is added at constant pressure, we have

$$Q_P = C_P \Delta T = \Delta U + W = \Delta U + P \Delta V$$

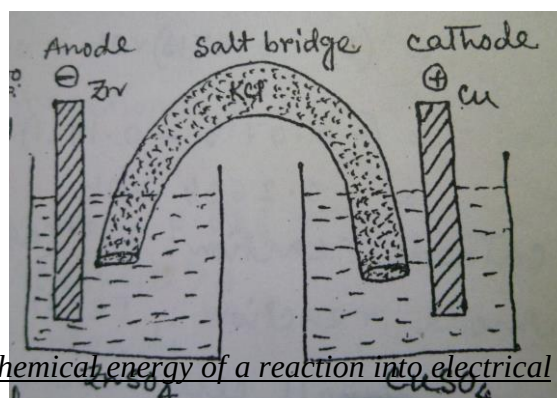
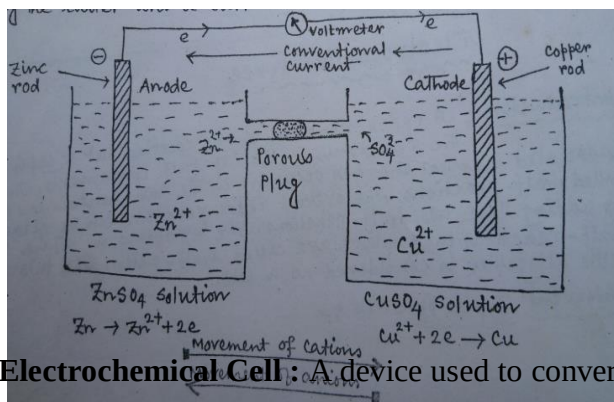
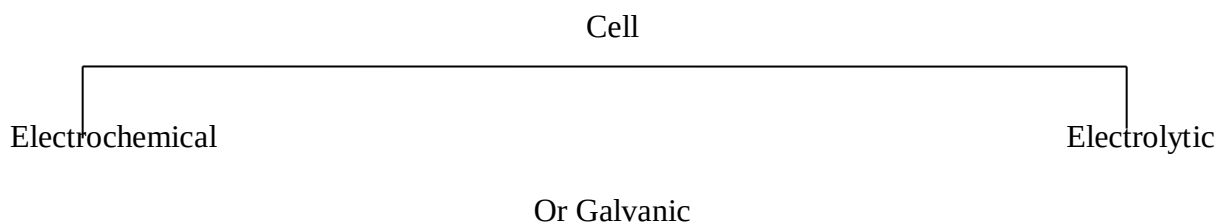
Relation between C_p and C_v

$$C_P - C_V = R$$

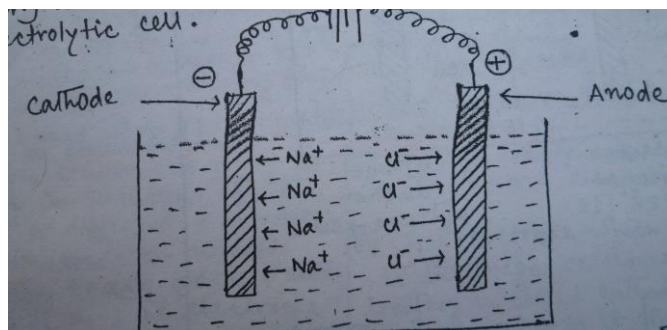
ELECTROCHEMICAL CELL

A redox reaction involves the transfer of electrons from one chemical species to another. The energy from a redox reaction can be used to accomplish work by constructing an electrochemical cell. In an electrochemical cell, the oxidation process and the reduction process are separated into two **half-cells** connected by an external wire. The half-cell with the oxidation process is losing negative charge (e^- loss) while the half-cell with the reduction process is gaining negative charge

(e^- gain). To maintain electrical neutrality in both half-cells, a salt bridge (or semipermeable membrane) must connect the two half-cells to permit the transfer of ions between the two solutions. Thus, the salt bridge completes the electrical circuit between the half cells.



Electrochemical Cell: A device used to convert chemical energy of a reaction into electrical energy is called Electrochemical Cell or Galvanic Cell.



Electrolytic Cell : A device used to convert electrical energy of a reaction into chemical energy is called Electrochemical Cell or Galvanic Cell.

The differences between Voltaic Cell and Electrolytic Cell

Characteristics	Voltaic Cell	Electrolytic Cell
Energy Change	chemical energy $\rightarrow$ electrical energy	electrical energy $\rightarrow$ chemical energy
Cathode and Anode	cathode : (+) terminal anode : (-) terminal	cathode : (-) terminal anode : (+) terminal
Flow of Electrons	electron flows from the negative electrode (anode) to the positive electrode (cathode)	electron flows from the positive electrode (anode) to the negative electrode (cathode)
Negative Terminal	electrons are released	cation receives electrons
Positive Terminal	electrons are received	anions release electrons

Electromotive force, also called **emf** (measured in volts), is the voltage developed by any source of electrical energy such as a battery or dynamo. It is generally defined as the electrical potential for a source in a circuit. A device that supplies electrical energy is called a seat of electromotive force or **emf**. Emfs convert chemical, mechanical, and other forms of energy into electrical energy. The product of such a device is also known as **emf**.

Standard cell potentials (E°_{cell}) can be calculated from potentials (E°) for oxidation and reductions reactions measured under standard conditions (1.00 M solutions, 1.00 atm pressure, 298 K). A positive cell potential means the reaction proceeds spontaneously in the direction the reaction is written; a reaction with a negative cell potential proceeds spontaneously in the reverse direction.

Module III: Organic Chemistry

Electronic factors that influence organic reactions include the **inductive effect**, the **electromeric effect**, **resonance effects**, and **hyperconjugation**. These electronic factors work off of organic molecules, which are themselves diverse and robust, despite their limited number of components. Most biological molecules are made from a combination of the following six elements: carbon, hydrogen, nitrogen, oxygen, phosphorus, and sulfur. Yet, the limited number of building blocks does not prevent organic compounds from taking on diverse properties in their physical characteristics and chemical reactivity.

Inductive Effect:

Definition: The inductive effect is the effect charge of a chemical bond has on the orientation on the adjacent bonds in a molecule. Electron clouds in a bond tend to orient themselves towards the more electronegative atom involved in the bond.

Types of inductive effect :

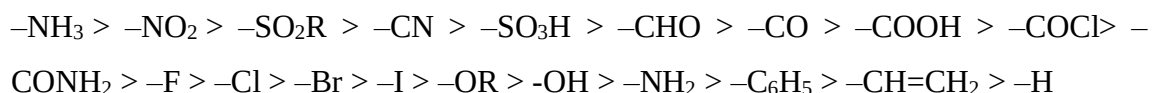
If the electronegative atom is then joined to a chain of atoms, usually carbon, the positive charge is relayed to the other atoms in the chain. This is the electron-withdrawing inductive effect, also known as the **-I - effect**.

However, Some groups, such as the alkyl group, are less electron-withdrawing than hydrogen and are therefore considered as electron-releasing. This is electron-releasing character and is indicated by the **+ I - effect**. In short, alkyl groups tend to give electrons, leading to induction effect.

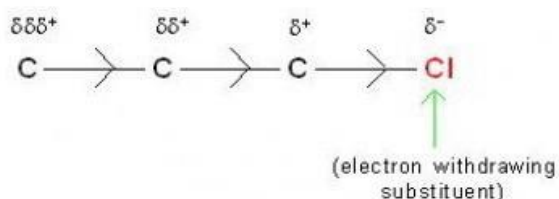
As the induced change in polarity is less than the original polarity, the inductive effect rapidly dies out and is significant only over a short distance. Moreover, the inductive effect is permanent but feeble since it involves the shift of strongly held σ -bond electrons and other stronger factors may overshadow this effect.

Relative inductive effects :

Relative inductive effects have been experimentally measured with reference to hydrogen, in decreasing order of -I effect or increasing order of +I effect, as follows:



Inductive effect in propyl chloride



Resonance Effect:

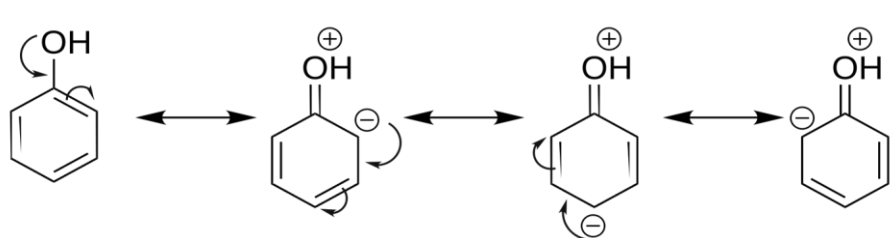
In chemistry, resonance or mesomerism is a way of describing delocalized electrons within certain molecules or polyatomic ions where the bonding cannot be expressed by one single Lewis structure. A molecule or ion with such delocalized electrons is represented by several contributing structures

Each contributing structure can be represented by a Lewis structure, with only an integer number of covalent bonds between each pair of atoms within the structure. Several Lewis structures are used collectively to describe the actual molecular structure, which is an approximate intermediate between the canonical forms called a resonance hybrid.

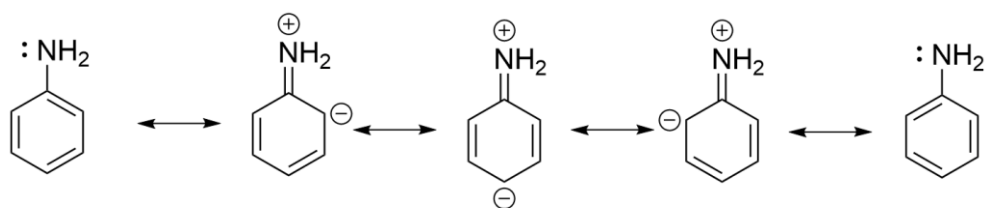
+R – effect

Groups or atoms that donate electrons through resonance exhibit +R – effect.

1. Resonance in phenol



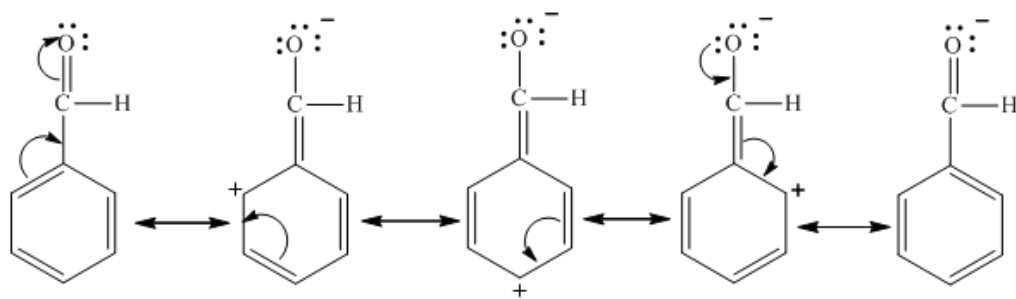
2. Resonance in aniline



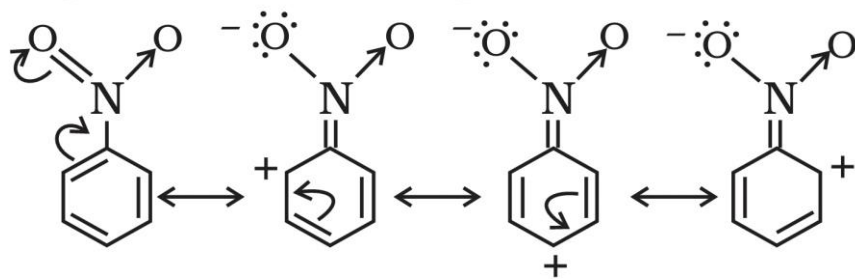
-R – effect

Groups or atoms that withdraws electrons through resonance exhibit -R – effect.

1. Resonance in benzaldehyde

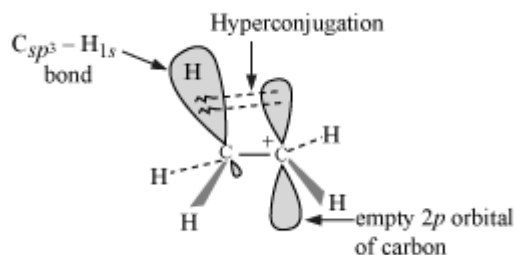


2. Resonance in nitrobenzene



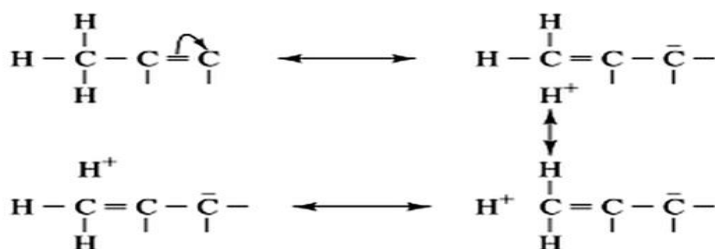
Hyperconjugation:

In organic chemistry, **hyperconjugation** is the interaction of the electrons in a sigma bond (usually C–H or C–C) with an adjacent empty (or partially filled) non-bonding p-orbital, antibonding σ or π orbital, or filled π orbital, to give an extended molecular orbital that increases the stability of the system.



In other words, it is the stabilising interaction that results from the interaction of the electrons in a σ -bond (usually **C–H** or **C–C**) with an adjacent empty or partially filled p-orbital or a π -orbital to give an extended molecular orbital that increases the stability of the system.

Example: Hyperconjugation In Ethyl Radical

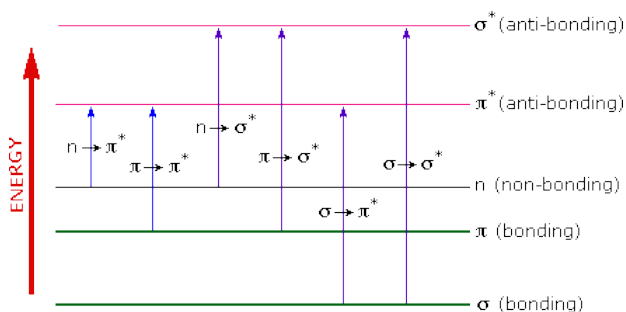


Spectroscopy

UV-Visible Absorption Spectra

To understand why some compounds are colored and others are not, and to determine the relationship of conjugation to color, we must make accurate measurements of light absorption at different wavelengths in and near the visible part of the spectrum. Commercial optical spectrometers enable such experiments to be conducted with ease, and usually survey both the near ultraviolet and visible portions of the spectrum. The visible region of the spectrum comprises photon energies of 36 to 72 kcal/mole, and the near ultraviolet region, out to 200 nm, extends this energy range to 143 kcal/mole. Ultraviolet radiation having wavelengths less than 200 nm is difficult to handle, and is seldom used as a routine tool for structural analysis.

The energies noted above are sufficient to promote or excite a molecular electron to a higher energy orbital. Consequently, absorption spectroscopy carried out in this region is sometimes called "electronic spectroscopy". A diagram showing the various kinds of electronic excitation that may occur in organic molecules is shown on the left. Of the six transitions outlined, only the two lowest energy ones (left-most, colored blue) are achieved by the energies available in the 200 to 800 nm spectrum. As a rule, energetically favored electron promotion will be from the **highest occupied molecular orbital (HOMO)** to the **lowest unoccupied molecular orbital (LUMO)**, and the resulting species is called an **excited state**.



When sample molecules are exposed to light having an energy that matches a possible electronic transition within the molecule, some of the light energy will be absorbed as the electron is promoted to a higher energy orbital. An optical spectrometer records the wavelengths at which absorption occurs, together with the degree of absorption at each wavelength. The resulting spectrum is presented as a graph of absorbance (A) versus wavelength, as in the isoprene spectrum shown below. Since isoprene is colorless, it does not absorb in the visible part of the spectrum and this region is not displayed on the graph. **Absorbance** usually ranges from 0 (no absorption) to 2 (99% absorption), and is precisely defined in context with spectrometer operation. Because the absorbance of a sample will be proportional to the number of absorbing molecules in the spectrometer light beam (e.g. their molar concentration in the sample tube), it is necessary to correct the absorbance value for this and other operational factors if the spectra of different compounds are to be compared in a meaningful way. The corrected absorption value is called "molar absorptivity", and is particularly useful when comparing the spectra of different compounds and determining the

relative strength of light absorbing functions (chromophores). **Molar absorptivity** (ϵ) is defined as:

$$\epsilon = A / c l$$

where, A= absorbance, c = sample concentration in moles/litre & l = length of light path through the sample in cm.

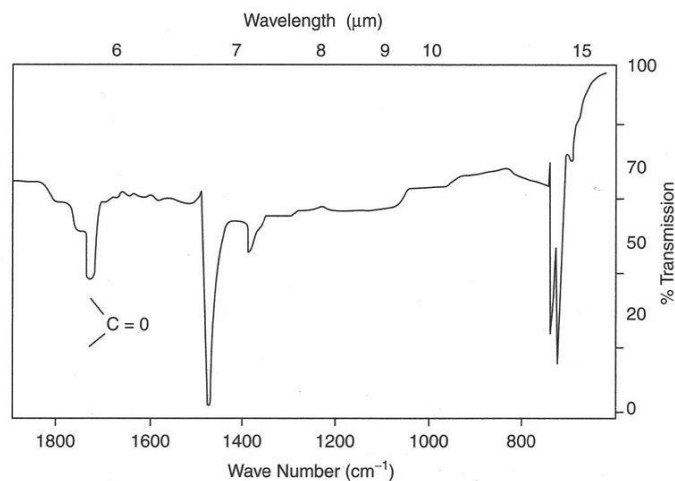
Infrared Spectroscopy

Infrared Spectroscopy is the analysis of infrared light interacting with a molecule. This can be analyzed in three ways by measuring absorption, emission and reflection. The main use of this technique is in organic and inorganic chemistry. It is used by chemists to determine functional groups in molecules. IR Spectroscopy measures the vibrations of atoms, and based on this it is possible to determine the functional groups. Generally, stronger bonds and light atoms will vibrate at a high stretching frequency (wavenumber).

Theory

Infrared (IR) spectroscopy is one of the most common and widely used spectroscopic techniques employed mainly by inorganic and organic chemists due to its usefulness in determining structures of compounds and identifying them. Chemical compounds have different chemical properties due to the presence of different functional groups.

Absorbing groups in the infrared region absorb within a certain wavelength region. The absorption peaks within this region are usually sharper when compared with absorption peaks from the ultraviolet and visible regions. In this way, IR spectroscopy can be very sensitive to determination of functional groups within a sample since different functional group absorbs different particular frequency of IR radiation. Also, each molecule has a characteristic spectrum often referred to as the fingerprint. A molecule can be identified by comparing its absorption peak to a data bank of spectra. IR spectroscopy is very useful in the identification and structure analysis of a variety of substances, including both organic and inorganic compounds. It can also be used for both qualitative and quantitative analysis of complex mixtures of similar compounds.



IR Spectrum of Formaldehyde